

A One-Pass, Collapse-Resistant Neural Topic Model for Mass2Motif Discovery

Abstract

Mass2Motif discovery requires a model to explain held-out spectral evidence while retaining a broad inventory of chemically interpretable fragment and neutral-loss patterns. We present a neural probabilistic topic model in which one sparse top-2 routing pass replaces iterative document-level inference. Fixed, train-only token features and learned topic prototypes define both the router and the topic-word decoder. Fragment and neutral-loss channels receive equal probability mass, while balanced assignments, positive-NPMI structure, and prototype separation address topic collapse. A validation-only seed-42 ablation removed unnecessary feature, view, reconstruction, initialization, and recycling mechanisms and replaced the context MLP with one linear map. We compare the resulting model with Tomotopy LDA using the same 1000 topics, seed 42, 6 CPU threads, scaffold split, and leakage-filtered MAG protocol. On validation, the neural model produces 853 optimized motifs and 234 useful high-confidence motifs, versus 607 and 138 for Tomotopy. On test, the corresponding useful-motif counts are 312 and 186. Tomotopy retains higher mean structure overlap score (SOS), so the result is a breadth-purity trade-off rather than uniform superiority. Recorded core fitting time is lower for the neural model (31.9 versus 76.0 minutes). This is a single-dataset, single-seed research comparison, not a claim that the neural model should replace the production backend.

1 Introduction

MS2LDA represents tandem mass spectra as documents, fragment and neutral-loss tokens as words, and recurrent co-occurrence patterns as Mass2Motifs [1]. A useful large-scale model must do more than minimize a predictive loss: it should retain many distinct motifs, assign spectra confidently enough for motif-level analysis, and support chemical annotation [2].

Conventional latent Dirichlet allocation (LDA) estimates a new document’s topic mixture by iterative local inference [7]. Amortized neural inference can replace that loop with a learned mapping, but a model with 1,000 topics can collapse: a small subset of topics absorbs most probability mass, several prototypes become near-duplicates, or learned topics ignore observed token co-occurrence. Mass-spectrometry vocabularies introduce another bias. If fragment and neutral-loss channel evidence is summed, the larger channel gains mass simply because it contains more tokens, even when its mean logit is no better.

We address these problems with one shared prototype geometry. It defines a fragment/loss-balanced topic-word distribution and a contextual top-2 router whose aggregated assignments form the document mixture in one pass. The training objective combines full-spectrum likelihood with safeguards against usage, geometric, and co-occurrence collapse. This report states the complete selected model and evaluation protocol directly.

2 Methods

2.1 Notation and data representation

Let $d \in \{1, \dots, D\}$ index spectra, $w \in \{1, \dots, V\}$ index the shared fragment/loss vocabulary, and $k \in \{1, \dots, K\}$ index topics. The count $c_{dw} \geq 0$ is the historical MS2LDA token multiplicity. The model returns a document-topic distribution $\theta_d \in \Delta^{K-1}$ and a topic-word distribution $\beta_k \in \Delta^{V-1}$, with

$$p(w \mid d) = \sum_{k=1}^K \theta_{dk} \beta_{kw}. \quad (1)$$

The public inputs are distributed in Zenodo record 20179680 [9]. The source MGF contains 41568 spectra; deterministic cleaning retains 38888. Molecules are keyed by the connectivity block of the InChIKey. Cyclic molecules are grouped by non-chiral Bemis-Murcko scaffold [3], while each acyclic connectivity group remains intact. Groups are assigned by descending size to 70/10/20 train, validation, and test targets using seed 42. The resulting matrices contain 27222 training, 3889 validation, and 7777 test spectra, with no compound or scaffold overlap.

Peak intensities are divided by the spectrum maximum. Peaks outside $m/z \in (0, 2000]$ or normalized intensity $[0.01, 1]$ are discarded; at most 1,000 peaks are retained, and a spectrum must retain at least five. Each peak yields a fragment token and, when positive, a precursor-minus-peak neutral-loss token, both rounded to two decimal places. A token at normalized intensity I occurs $\text{round}(100I)$ times. The vocabulary is constructed only from training spectra, requires document frequency at least three, preserves first-seen order, and contains 21233 tokens.

For held-out completion, whole physical peak groups are divided deterministically 50/50. A fragment and its corresponding loss therefore cannot occur on opposite sides. The observed half is renormalized without consulting the withheld maximum. Model development uses only training and validation. Test evaluation is called once after validation evaluation and MAG scoring are complete.

2.2 Fixed token features

Each token has a fixed feature vector $x_w \in \mathbb{R}^{50}$. Its first 48 coordinates are skip-gram negative-sampling (SGNS) embeddings trained only on training documents [4]. For five epochs, eight count-weighted positive pairs are sampled per document; each positive pair uses five negatives drawn with probability proportional to corpus frequency raised to 0.75. Source embeddings are initialized uniformly on $[-0.5/48, 0.5/48]$ and context embeddings at zero. SparseAdam uses learning rate 0.01 and batches of 8,192 pairs. Source and context embeddings are averaged and normalized after the fifth epoch. Two final one-hot coordinates mark fragment versus neutral loss. The normalized SGNS vector and type indicators are concatenated and normalized again. Token-mass Fourier features, molecular labels, and pretrained spectrum encoders are not used.

2.3 Shared topic geometry and balanced decoder

A learned bias-free matrix $W \in \mathbb{R}^{128 \times 50}$ projects the fixed features, and every topic has a learned prototype $q_k \in \mathbb{R}^{128}$:

$$z_w = \text{normalize}(Wx_w), \quad \hat{q}_k = \text{normalize}(q_k). \quad (2)$$

The matrix W is initialized orthogonally. The same z_w and $\hat{q}_k$ are used by both decoder and router. Decoder logits are temperature-scaled cosine similarities,

$$\ell_{kw} = \frac{2\hat{q}_k^\top z_w}{\tau_\beta}, \quad \tau_\beta = 0.18. \quad (3)$$

Let V_F and V_L denote the fragment and neutral-loss vocabularies. Each channel receives exactly one half of the topic probability mass, and a softmax within that channel determines token rankings:

$$\beta_{kw} = \frac{1}{2} \frac{\exp(\ell_{kw})}{\sum_{v \in V_{t(w)}} \exp(\ell_{kv})}. \quad (4)$$

Thus $\sum_w \beta_{kw} = 1$, vocabulary size cannot change channel mass, rankings within each channel are unchanged, and the channel correction adds no learned parameter.

2.4 Contextual top-2 router and document mixture

For spectrum d , define the count-weighted embedding sum $s_d = \sum_w c_{dw} z_w$. For each non-zero token entry, the leave-one-out context is

$$\bar{z}_{d \setminus w} = \frac{s_d - c_{dw} z_w}{\max(\sum_v c_{dv} - c_{dw}, 1)}. \quad (5)$$

A learned bias-free matrix $C \in \mathbb{R}^{128 \times 128}$ gives

$$h_{dw} = \text{normalize}(z_w + C\bar{z}_{d \setminus w}), \quad u_d = \text{normalize}(s_d). \quad (6)$$

C is initialized to the identity, so the initial correction is the normalized leave-one-out context itself. Local and document evidence are added before routing:

$$a_{dwk} = h_{dw}^\top \hat{q}_k + u_d^\top \hat{q}_k, \quad r_{dw}^{\text{soft}} = \text{softmax}_k(a_{dwk}/T). \quad (7)$$

Only the two largest entries of r_{dw}^{soft} are retained and renormalized to form r_{dw}^{hard} . Training uses the straight-through quantity $r^{\text{hard}} + r^{\text{soft}} - \text{stopgrad}(r^{\text{soft}})$; inference uses r^{hard} directly.

Count-weighted routed mass is $M_{dk} = \sum_w c_{dw} r_{dwk}$. The same whole-spectrum score supplies a multiplicative document gate,

$$g_{dk} = \text{softmax}_k(u_d^\top \hat{q}_k/T), \quad \theta_{dk} = \frac{M_{dk} g_{dk}^\gamma}{\sum_j M_{dj} g_{dj}^\gamma}, \quad \gamma = 0.75. \quad (8)$$

The gate is detached in Equation 8 during training, so it cannot reduce the loss by sharpening itself; its document score still learns through Equation 7. Multiplication preserves exact zero support from token routing. If a spectrum has no routed mass, θ_d is explicitly uniform. Equations 7–8 require one forward pass and no per-spectrum optimization loop.

The count-weighted completion loss implements Equation 1 exactly at non-zero target entries:

$$\mathcal{L}_{\text{comp}}(\theta, X) = -\frac{1}{\sum_{dw} c_{dw}} \sum_{dw} c_{dw} \log \left(\sum_k \theta_{dk} \beta_{kw} \right). \quad (9)$$

Sparse indexing avoids materializing the dense matrix $\theta\beta$ but does not change its probabilities.

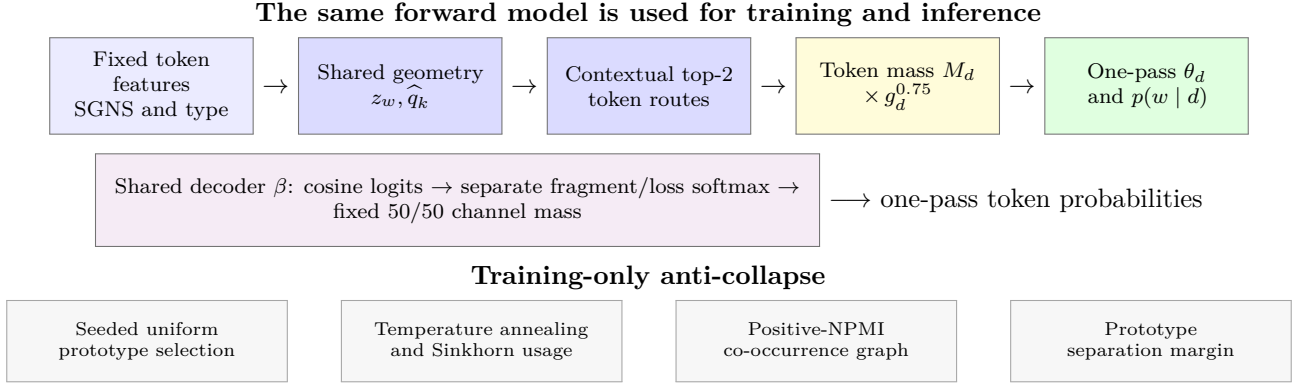


Figure 1: The implemented model. Fixed train-only features and learned prototypes define both the one-pass router and balanced decoder. The lower row contains training-only safeguards and is absent from inference.

2.5 Alternating full-spectrum objective

Optimization alternates two gradient blocks using one AdamW optimizer. In the *router block*, β is cached and detached. One full-spectrum route is optimized with likelihood, Sinkhorn targets, and prototype separation:

$$\mathcal{L}_{\text{router}} = \mathcal{L}_{\text{comp}} + \omega_S \mathcal{L}_{\text{Sinkhorn}} + 5\mathcal{L}_{\text{sep}}. \quad (10)$$

For a mini-batch with N routed entries, log-space Sinkhorn iteration projects $\exp(a/\varepsilon)$, with $\varepsilon = 0.05$, to detached targets Q whose rows sum to one and whose columns sum to N/K [5, 6]. $\mathcal{L}_{\text{Sinkhorn}}$ is cross-entropy from Q to $\text{softmax}(a/T)$. Sinkhorn is a training target only; it is not part of held-out inference.

In the *topic block*, routes are recomputed without gradients and held fixed. Decoder geometry is then optimized with

$$\mathcal{L}_{\text{topic}} = \mathcal{L}_{\text{comp}} + \mathcal{L}_{\text{NPMI}} + 5\mathcal{L}_{\text{sep}}, \quad (11)$$

using the same full training spectra. A full router pass with batch size 128 is followed by four topic-block updates with batch size 512 per epoch.

2.6 Anti-collapse design

The safeguards target different collapse modes; removing one is not equivalent to strengthening another.

1. **Deterministic initialization.** Seed 42 selects 1000 distinct projected training tokens uniformly without replacement. No clustering or frequency weighting is used.
2. **Annealed sparse routing.** Routing temperature falls linearly from 0.5 to 0.1 over 30 epochs. Early soft assignments permit exploration; later top-2 assignments specialize.
3. **Balanced usage.** The Sinkhorn coefficient is 0.25 for the first ten epochs and then decreases linearly to 0.05 at epoch 40. It resists early winner-take-all collapse without forcing uniform usage at the end of training.
4. **Train-only co-occurrence structure.** Binary training documents define a positive-NPMI graph. Tokens require document frequency at least ten, pair frequency at least three, and mutual membership among at most 16 strongest neighbours. If A is the resulting weighted sparse adjacency, the retained edge weight is

$$A_{wv} = \frac{\log[p(w, v)/(p(w)p(v))]}{-\log p(w, v)}, \quad A_{wv} > 0, \quad (12)$$

where probabilities are empirical document frequencies. Only mutual top-neighbour edges remain. The training loss is

$$\mathcal{L}_{\text{NPMI}} = -\frac{1}{K} \sum_k \log \left(\sum_{wv} \beta_{kw} A_{wv} \beta_{kv} \right). \quad (13)$$

This favours probability mass on training-supported token pairs.

5. **Prototype separation.** For every prototype, the eight nearest other prototypes are penalized above cosine margin 0.3:

$$\mathcal{L}_{\text{sep}} = \text{mean}_{k,j} \left[\max(0, \hat{q}_k^\top \hat{q}_j - 0.3) \right]^2. \quad (14)$$

This detects geometric duplication that usage balance cannot detect.

6. **Numerical controls.** PyTorch deterministic algorithms are required under the six-thread allowance. Non-finite losses or gradients stop training immediately, and gradient norm is clipped to 5.0. Training always runs to the predeclared epoch 40; no validation-metric model search is performed.

AdamW uses learning rate 10^{-3} , weight decay 10^{-4} , and seed 42. Table 3 lists the complete settings.

2.7 Validation-only architecture selection

The committed seed-42 model was loaded and its validation likelihood and MAG/SOS values were reproduced without retraining. Components were then removed one at a time, with every candidate trained from the same deterministic initialization. Test matrices and records were unavailable to candidate runs. The selected model removes view consistency, local reconstruction, dead-topic recycling, Fourier coordinates, paired views, weighted k -means++ seeding, and adaptive channel mass, and replaces the context MLP with the linear map C . It retains the document evidence and gate, top-2 routing, Sinkhorn targets, positive-NPMI loss, prototype separation, and alternating optimization. The first lock replay exposed nondeterministic parallel gradients, so strict deterministic PyTorch execution was enabled and U7 was trained again as a corrected candidate before an independent lock confirmation. The selection was frozen before one final test evaluation. The complete ordered validation evidence is stored in the accompanying ablation ledger.

2.8 Comparator and evaluation

Tomotopy 0.13.0 [8] is trained independently on the same training matrix and never acts as a teacher. It uses $K = 1000$, scalar initial $\alpha = 0.6$ with optimization interval 10, $\eta = 0.1$, seed 42, 6 workers, and parallel scheme 3. Training advances in blocks of 50 to at most 5,000 iterations and stops when all of the last ten relative perplexity changes are below 0.005. The recorded fit stops at 750 iterations. Held-out Tomotopy mixtures use 100 iterations, 6 workers, parallel scheme 1, and independent-document inference. The neural model has no corresponding iteration count because its document mapping is amortized into one routing pass.

Reported fitting time is optimization-loop wall time. For the neural model it excludes MGF processing, SGNS feature construction, co-occurrence graph construction, initialization, and evaluation; for Tomotopy it excludes sparse matrix expansion, model construction, and evaluation. This gives like-for-like core fitting time but is not total workflow time.

Predictive evaluation is count-weighted completion NLL per in-vocabulary token. Chemical evaluation follows the paper-aligned MAG workflow [2]. Before retrieval, all reference entries sharing a validation or test connectivity key are removed. The 20 most probable tokens of each topic form a motif spectrum. MAG retrieves up to 1,000 neighbours, retains ten unique molecules, clusters at cosine similarity 0.9, and optimizes motif features. A topic with at least one optimized feature counts toward annotation coverage.

A high-confidence topic-spectrum association uses the full-spectrum mixture and requires $\theta_{dk} \geq 0.5$. For an annotated topic, MACCS bits present in at least 80% of its clustered MAG molecules define binary fingerprint A_k . For associated compound i with fingerprint B_i , the containment score is

$$s_{ki} = \frac{|A_k \cap B_i|}{|A_k|}, \quad \text{SOS}_k = \frac{1}{|C_k|} \sum_{i \in C_k} s_{ki}, \quad (15)$$

where C_k contains unique connectivity keys, preventing repeated spectra of one molecule from dominating. A topic is SOS-evaluable only when MAG retained at least one optimized feature and C_k contains an associated molecule with a valid fingerprint. Among SOS-evaluable high-confidence motifs, bands are high (> 0.8), intermediate ($0.6 \leq \text{SOS} \leq 0.8$), and low (< 0.6). A *useful high-confidence motif* lies in the high or intermediate band. Optimized, evaluable, and useful motif counts, SOS bands, mean/median SOS, and fitting time form the primary comparison. NLL, collapse checks, and warm resident inference are secondary diagnostics.

3 Results

3.1 Validation comparison

Table 1: Primary validation comparison.

Metric	Neural	Tomotopy
Optimized motifs	853	607
MAG annotation coverage	85.3%	60.7%
High-confidence SOS-evaluable motifs	396	206
SOS high (> 0.8)	64	58
SOS intermediate ($0.6-0.8$)	170	80
SOS low (< 0.6)	162	68
Useful motifs (high-confidence; $\text{SOS} \geq 0.6$)	234	138
Mean SOS (high-confidence)	0.6367	0.6761
Median SOS (high-confidence)	0.6324	0.6854
Model-fitting time (minutes)	31.9	76.0

The neural model yields 853 optimized motifs versus 607 for Tomotopy, and 396 high-confidence SOS-evaluable motifs versus 206. Useful-motif counts are 234 and 138, respectively. The broader neural inventory has lower mean SOS (0.6367 versus 0.6761). Recorded core fitting time is lower: 31.9 minutes for neural MS2LDA and 76.0 minutes for Tomotopy.

3.2 Test confirmation

Table 2: Primary test comparison after validation-only development.

Metric	Neural	Tomotopy
Optimized motifs	853	607
MAG annotation coverage	85.3%	60.7%
High-confidence SOS-evaluable motifs	547	333
SOS high (> 0.8)	84	66
SOS intermediate ($0.6-0.8$)	228	120
SOS low (< 0.6)	235	147
Useful motifs (high-confidence; $\text{SOS} \geq 0.6$)	312	186
Mean SOS (high-confidence)	0.6366	0.6369
Median SOS (high-confidence)	0.6281	0.6145
Model-fitting time (minutes)	31.9	76.0

The test result preserves the validation pattern. Neural MS2LDA yields 312 useful high-confidence motifs versus 186, and 547 SOS-evaluable motifs versus 333. Annotation coverage is 85.3% versus 60.7%. Mean SOS remains lower for the neural model (0.6366 versus 0.6369); the corresponding medians are reported in Table 2.

4 Discussion

The fixed 50/50 channel decoder resolves vocabulary-size bias without estimating a channel-mass mechanism. It is both simpler and easier to interpret than the original adaptive blend, while retaining within-channel cosine rankings.

Parity is metric-dependent. The neural model produces more optimized, high-confidence evaluable, and useful motifs, predicts withheld tokens with lower NLL (Appendix C), and has lower recorded core fitting time. Tomotopy has higher mean validation and test SOS. The neural result is therefore broader, not uniformly purer. A single seed was used deliberately to improve and inspect one architecture without averaging away its failure modes, but this design does not measure uncertainty across initializations or datasets.

The most consequential future extension is a jointly trained spectrum-level encoder. The current u_d is a normalized sum of projected token embeddings; a set or transformer encoder could represent global peak relations that a sum cannot, while the prototype decoder retains interpretable motifs. Other natural extensions are sparse chemical supervision where reliable labels exist, fine-tuning on a new dataset, GPU-native sparse batches, and optionally freezing known motif prototypes while other topics learn. Selective freezing is useful but is not unique to neural inference because variational-Bayes LDA can also fix chosen motifs. None of these extensions contributes to the reported result.

The limitations are concrete. This is one dataset and one seed. MAG outcomes depend on the retrieval library, clustering, and motif optimization. Warm inference excludes parsing, feature construction, model

loading, and cold-start costs. Finally, topic counts and SOS characterize the evaluable motif inventory; they do not prove that every topic corresponds to a distinct biochemical substructure.

5 Conclusion

A shared prototype geometry, one-pass contextual routing, fixed equal fragment/loss mass, and complementary anti-collapse training produce a broad neural Mass2Motif inventory with strong predictive fit. Under the fixed seed-42, $K = 1000$, 6-thread protocol, neural MS2LDA exceeds Tomotopy in optimized and useful motif counts, while Tomotopy retains higher mean SOS. The next scientific step is replication across datasets and seeds, not further adaptation to the evaluated test split.

A Exact hyperparameters

Table 3: Values generated from `protocol.json`.

Quantity	Value
Seed; topics; CPU threads	42; 1000; 6
Token features	48 SGNS + 2 type = 50 dimensions
SGNS	SparseAdam; 5 epochs; 8 pairs/document; 5 negatives; power 0.75; batch 8192; lr 0.01
Token projection; context projection	128; identity-initialized bias-free linear 128 by 128
Decoder	temperature 0.18; separate channel softmax; fixed 50/50 fragment/loss mass
Router	top-2; additive document evidence; gate exponent 0.75
Training spectra	one full-spectrum path; no corrupted views
AdamW	lr 0.001; weight decay 0.0001; gradient clip 5.0; deterministic PyTorch algorithms
Batches and epochs	router 128; topic 512; 4 topic updates/epoch; 40 epochs
Loss weights	NPMI 1.0; separation 5.0
Routing temperature	0.5 to 0.1 over 30 epochs
Sinkhorn	epsilon 0.05; 5 iterations; weight 0.25 to 0.05
NPMI graph	min DF 10; min pair 3; 16 neighbours; positive NPMI
Prototype separation	8 neighbours; margin 0.3
Tomotopy	alpha 0.6; eta 0.1; parallel 3; step 50; max 5000; stop 10 changes < 0.005; inference 100
MAG/SOS	top 20 motif tokens; search 1000; 10 unique molecules; membership ≥ 0.5 ; cluster cosine 0.9; fingerprint consensus 0.8

B Equation–implementation correspondence

Table 4: Direct map from the mathematical description to executable code.

Report component	Implementation
Token features x_w	<code>data.py: build_token_features</code>
Projected tokens z_w	<code>model.py: projected_tokens</code>
Decoder β	<code>model.py: topic_word_distribution</code>
Leave-one-out context	<code>model.py: _route_embeddings</code>
Top-2 routing	<code>model.py: route</code>
Gated θ_d	<code>model.py: aggregate_theta</code>
Likelihood and regularizers	<code>objectives.py</code>
Alternating optimization	<code>training.py; optimization.py</code>
Held-out inference	<code>evaluation.py</code>
MAG and SOS	<code>mag.py; chemical.py</code>

C Secondary diagnostics and practical inference

On test, 304 topics exceed uniform mean corpus usage and the median entropy-derived effective topic count is 6.68 per spectrum. These quantities are collapse checks, not headline chemical outcomes. Completion NLL per token is 9.1221 versus 9.6622 on validation and 9.1358 versus 9.7569 on test (neural versus Tomotopy).

In warm in-memory inference on resident models and batches of 128 spectra, the neural router processes 10828.1 spectra/s versus 30.0 spectra/s for Tomotopy, a 360.3-fold ratio. Both measurements use 6 CPU threads; the neural model uses one routing pass and Tomotopy uses 100 inference iterations. This result is relevant to repeated in-memory batches or a resident service, not end-to-end application latency.

D Executable reproduction

The repository contains one current protocol and one current model artifact. The latter consists only of the model state, vocabulary, and the two inference temperatures. From the repository root:

```
conda env create -f environment-neural-ms2lda.yml
conda env create -f environment-msnlib-mag.yml
conda run -n ms2lda-neural python \
  scripts/download_msnlib_validation_assets.py \
  --data-root /path/to/MSnLib-assets
conda run -n ms2lda-neural python -m benchmarks.neural_ms2lda run \
  --data-root /path/to/MSnLib-assets --run /path/to/run
python scripts/generate_neural_ms2lda_report.py
```

The acquisition command validates the public source data before training. Validation evaluation and MAG scoring finish before test evaluation is invoked. The comparator is trained from the same training matrix, so no external Tomotopy result is required.

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